Data Structures and Algorithms -3 (25CS2103E)

Intelligent Resume Search and Candidate Ranking System

Team Members

Gnaneshwar	2520030552
Yashas	2520030401
Charan	2520030321

Under the Esteemed Guidance of

Dr.V. Sireesha

Assistant professor

Contents

Abstract

TalentFlow is a Java-based recruiter-side resume searching and candidate ranking application. The system allows recruiters to upload candidate resumes in PDF format, extract useful candidate information, search for required skills or keywords, handle spelling variations, and rank candidates according to their relevance.

The project mainly focuses on the application of Data Structures and Algorithms in a real-world recruitment problem. The major algorithms used in the system are KMP (Knuth–Morris–Pratt) for efficient pattern searching, Edit Distance for handling spelling mistakes and variations, and Candidate Matching and Ranking for identifying the most relevant candidates.

The project combines JavaFX for the user interface, Apache PDFBox for resume processing, and DSA algorithms for efficient candidate searching and ranking.

Introduction

Recruiters may need to examine a large number of resumes to identify candidates who possess specific skills and qualifications. Manual resume screening can be time-consuming and repetitive.
Searching resumes using simple keyword matching may also fail when the recruiter enters a spelling variation or when the information is present in a slightly different form.
TalentFlow is proposed as a solution to this problem. The system provides a recruiter-oriented application where resumes can be uploaded and processed. The extracted candidate information is then searched using DSA-based algorithms.

The basic working process is:
Resume Upload → PDF Text Extraction → Candidate Information → Search Query → KMP Pattern Matching → Edit Distance → Candidate Matching → Candidate Ranking → Ranked Results.

Literature Survey

Existing resume screening approaches commonly use manual review or basic keyword-based searching.

Manual screening requires recruiters to inspect resumes individually, which becomes difficult when the number of resumes increases. Basic keyword searching can also have limitations because it generally depends on exact matching. A spelling mistake in a recruiter search may prevent the correct candidate from being identified.

TalentFlow addresses these limitations by combining efficient pattern searching and approximate string matching.

The major approaches considered are manual resume screening, basic keyword search, KMP pattern searching, Edit Distance, and candidate ranking. KMP provides efficient pattern searching, Edit Distance handles spelling differences, and candidate ranking helps recruiters identify highly relevant candidates first.

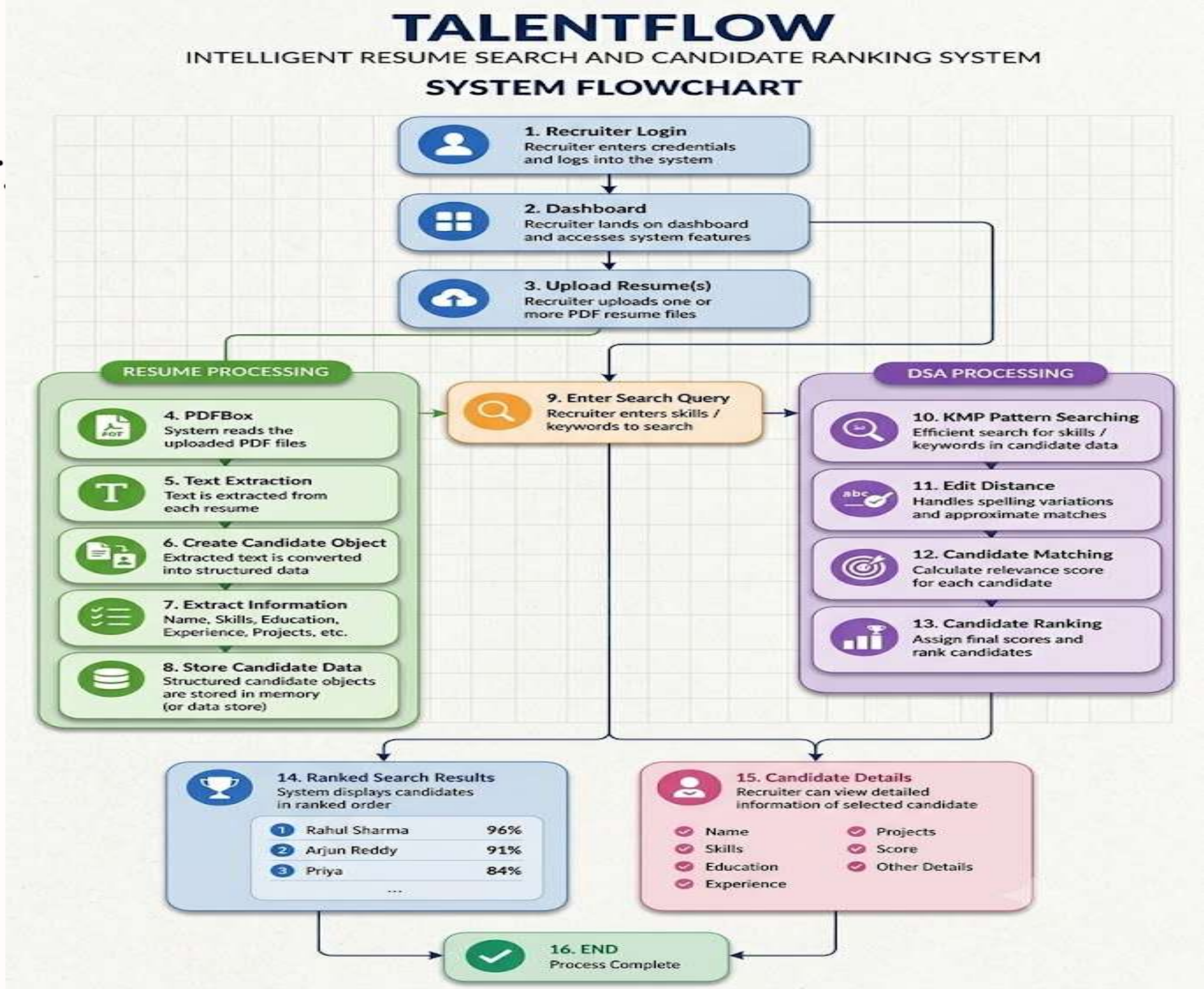
Problem Statement and Objectives

To develop a Java-based recruiter-side resume searching and candidate ranking system that efficiently searches candidate information, handles spelling variations, and ranks candidates according to their relevance using Data Structures and Algorithms.

The main objectives of TalentFlow are:

1. To develop a user-friendly recruiter-side application using JavaFX.
2. To allow recruiters to upload candidate resume PDF files.
3. To extract useful information from resumes.
4. To create structured Candidate objects containing name, skills, education, experience, and projects.
5. To implement KMP for efficient pattern and skill searching.

Flow Chart


TALENTFLOW
INTELLIGENT RESUME SEARCH AND CANDIDATE RANKING SYSTEM
SYSTEM FLOWCHART
1. Recruiter Login
Recruiter enters credentials and logs into the system
2. Dashboard
Recruiter lands on dashboard and accesses system features
3. Upload Resume(s)
Recruiter uploads one or more PDF resume files
RESUME PROCESSING
4. PDFBox
System reads the uploaded PDF files
5. Text Extraction
Text is extracted from each resume
6. Create Candidate Object
Extracted text is converted into structured data
7. Extract Information
Name, Skills, Education, Experience, Projects, etc.
8. Store Candidate Data
Structured candidate objects are stored in memory (or data store)
9. Enter Search Query
Recruiter enters skills / keywords to search
DSA PROCESSING
10. KMP Pattern Searching
Efficient search for skills / keywords in candidate data
11. Edit Distance
Handles spelling variations and approximate matches
12. Candidate Matching
Calculate relevance score for each candidate
13. Candidate Ranking
Assign final scores and rank candidates
14. Ranked Search Results
System displays candidates in ranked order
1 Rahul Sharma 96%
2 Arjun Reddy 91%
3 Priya 84%
...
15. Candidate Details
Recruiter can view detailed information of selected candidate
Name
Skills
Education
Experience
Projects
Score
Other Details
16. END
Process Complete

Results

The TalentFlow system provides a recruiter-oriented platform for searching and ranking candidates based on their resume information.

The implemented system is designed to process resume PDF files using Apache PDFBox and convert the extracted information into structured Candidate objects. The candidate data contains important details such as name, skills, education, experience, and projects.

The DSA module performs candidate searching and matching using KMP and Edit Distance. KMP is used for efficient searching of required skills and keywords, while Edit Distance helps identify approximate matches when spelling variations occur.

After the matching process, candidates are assigned relevance scores and displayed according to their ranking.

For example:

Search Query: Java, SQL, DSA

Ranked Results:

1. Rahul Sharma – 96%
2. Arjun Reddy – 91%
3. Priya – 84%

Hardware and Software requirements

Hardware Requirements

The basic hardware requirements are:

Computer or Laptop

Minimum 4 GB RAM

Minimum 10 GB available storage

Keyboard

Mouse.

Software Requirements

The software requirements are:

Operating System: Windows / Linux

Programming Language: Java

Frontend: JavaFX

PDF Processing: Apache PDFBox

Build Tool: Maven

IDE: IntelliJ IDEA / Eclipse / VS Code

Version Control: Git / GitHub.

Conclusion

TalentFlow demonstrates the practical application of Data Structures and Algorithms in a recruitment-based problem.

The system combines JavaFX, PDF Processing, KMP, Edit Distance, and Candidate Ranking.

The application aims to reduce the effort required to search and compare resumes. KMP provides efficient pattern searching, while Edit Distance helps handle spelling variations. Candidate matching and ranking then help display the most relevant candidates first.

The modular structure allows the three major components—frontend, resume processing, and DSA algorithms—to be developed independently and integrated into the final application.

Future Scope

The future enhancements of TalentFlow can include:

1. Support for a larger number of resumes.
2. Improved resume information extraction.
3. Support for additional resume formats.
4. More advanced candidate scoring.
5. Additional filtering based on skills, education, experience, and projects.
6. More advanced DSA-based searching and ranking.
7. Improved performance for large datasets.
8. Cloud-based resume storage.
9. Authentication and role-based access.
10. More advanced candidate recommendation features.

Thank you